Ethan Austin-Cruse

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Education

Georgia Institute of Technology Atlanta, GA

Bachelor of Science in Computer Science, Minor in Mathematics *May 2028*

- **Relevant Coursework:** Computer Organization & Programming, Data Structures & Algorithms, Intro to Object Oriented Programming, Linear Algebra, Multivariable Calculus, Probability & Statistics

Experience

Conturo Feb. 2026 – Present

Founder

- Shipped a REST API in TypeScript using Hono, Zod, and PostgreSQL enabling nonprofit organizations to transfer 100k+ contacts with 10-20 identity fields each between conturo.io and their CRM, ESP, and analytics software
- Processed 1M+ contact datums monthly with cron jobs, Playwright, and OpenAI API to maintain B2C database
- Developed MCP server and UI/UX for AI chat interface allowing nonprofits to export contacts in natural language
- Built crash-safe export worker with atomic job claiming and lease-based retry covered by 200+ Deno tests

Apprentice to Google Vice President Jan. 2025 – Feb. 2025

Software and Machine Learning

- Designed chess engine using Alpha-Beta pruning with 500k–650k evals/sec motivated by benchmarking RL agent
- Researched gradient descent applications and cost functions in LLMs under Daniel Berlin, Google SWE VP

Center for Puppetry Arts Nov. 2024 – Nov. 2025

Data Engineer

- Aggregated 150k+ marketing data points with Python scripts to 5x revenue on new education segment sales
- Improved educational info collection efficiency by up to 100x utilizing Playwright and AI parsing on HTML

Projects

USB Relay Kernel Driver | *C, Linux Kernel, USB, Git* Oct. 2025 – Dec. 2025

- Built a Linux kernel USB driver in C exposing 8-channel relay boards as /dev character devices
- Implemented USB hot-plug via probe/disconnect, endpoint discovery, and bulk transfers with 1s timeouts
- Enabled FTDI bit-bang mode via vendor USB control requests, encoding 256 relay states in one byte
- Scaled to 4 boards (32 relays) with IDA allocation, per-device mutexes, and leak-free cleanup

Tempr | *TypeScript, React Native, Supabase, Gemini API* Feb. 2026

- Placed 2nd in GT Hacklytics Entertainment track, shipping a React Native app with Spotify Auth in 36 hrs
- Assembled multimodal Gemini pipelines turning text or photo embeddings into playlists with Spotify API
- Pushed Tinder-style short form music discovery feature for future vector similarity search recommendation engine

Athletics & Leadership

USA Ultimate National Team — Gold Medal Dec. 2025 – Jul. 2026

Defensive Specialist *Logroño, Spain*

- World champion at the U-20 WFDF Ultimate Frisbee Championships representing the United States delegation

Eagle Scout Aug. 2025

Scouting America, Troop 175 *Decatur, GA*

- Led 8 youth volunteers for a total of 75 hours to build mushroom inoculation tables increasing local produce yield

Technical Skills

Languages: TypeScript/JavaScript, Java, Python, C, SQL, HTML/CSS
Frameworks: React, Node.js, Hono, Matt-Pocock-Skills, Tailwind, Spring Boot
Developer Tools: Git, GitHub, CI/CD, Linux, Claude Code, Codex, PostgreSQL, Docker, Vercel, OpenAI API, Jev
Libraries: Zod, AI SDK, Playwright, Pandas, TensorFlow, Scikit-learn, Beautiful Soup, Radix